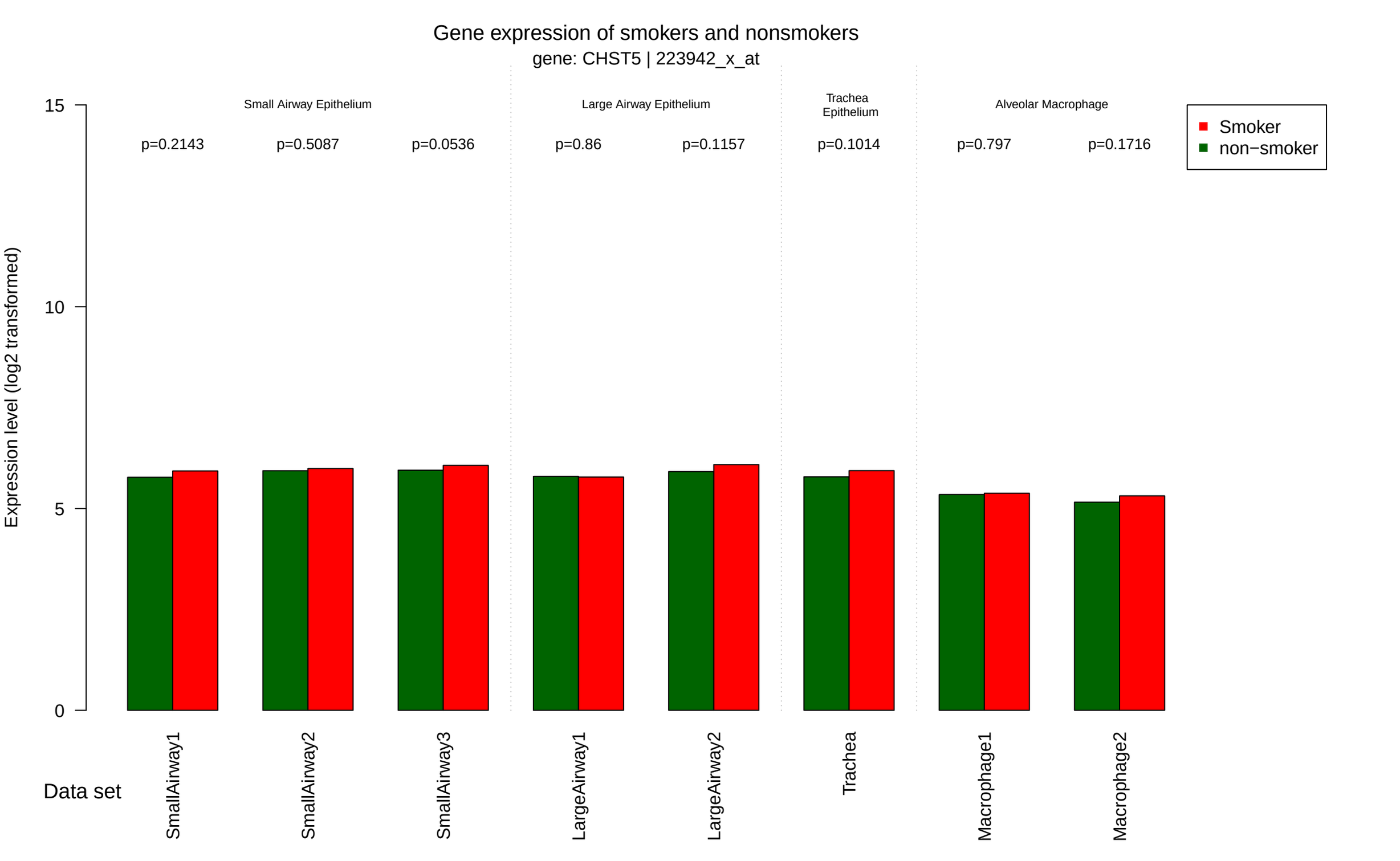
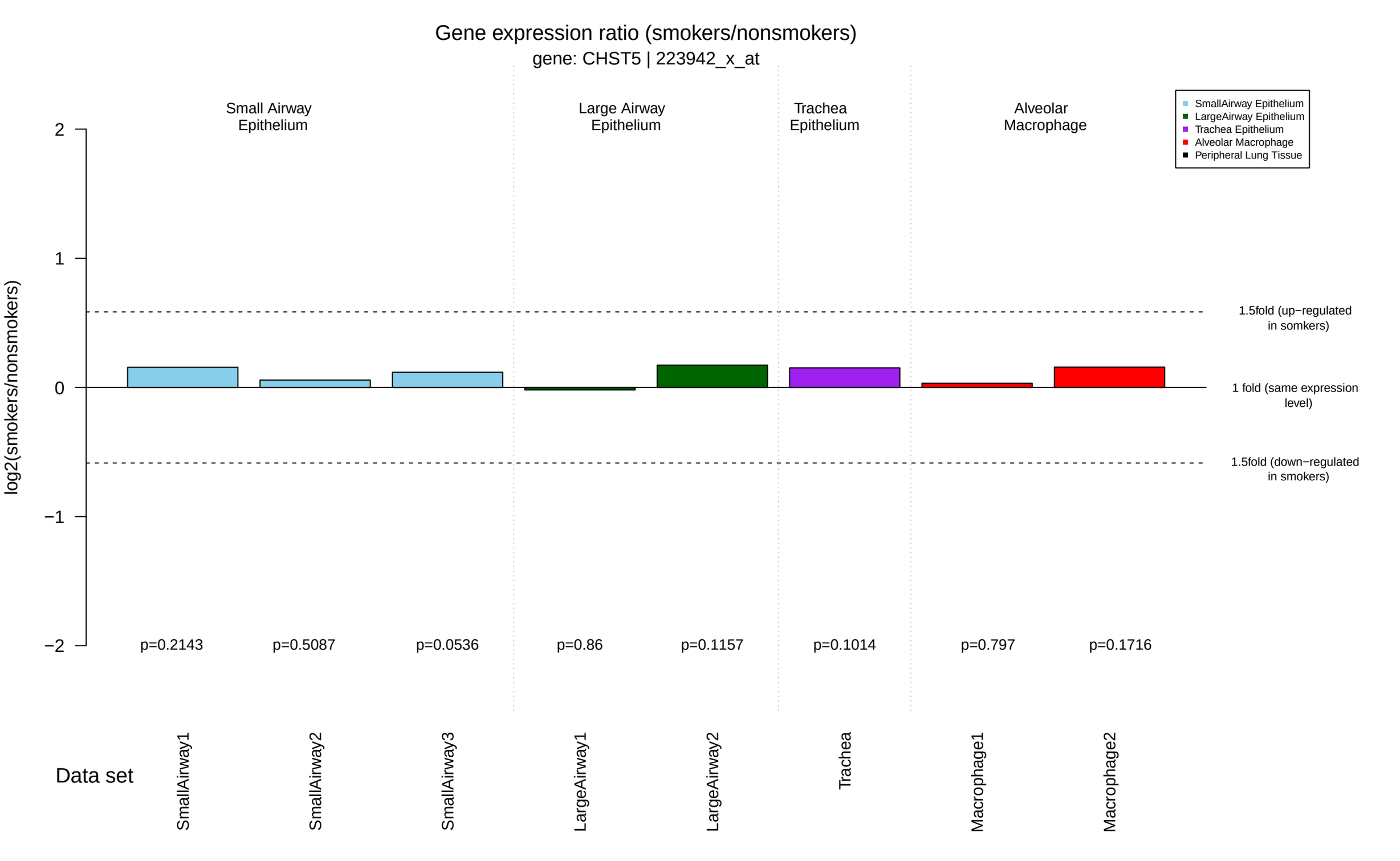
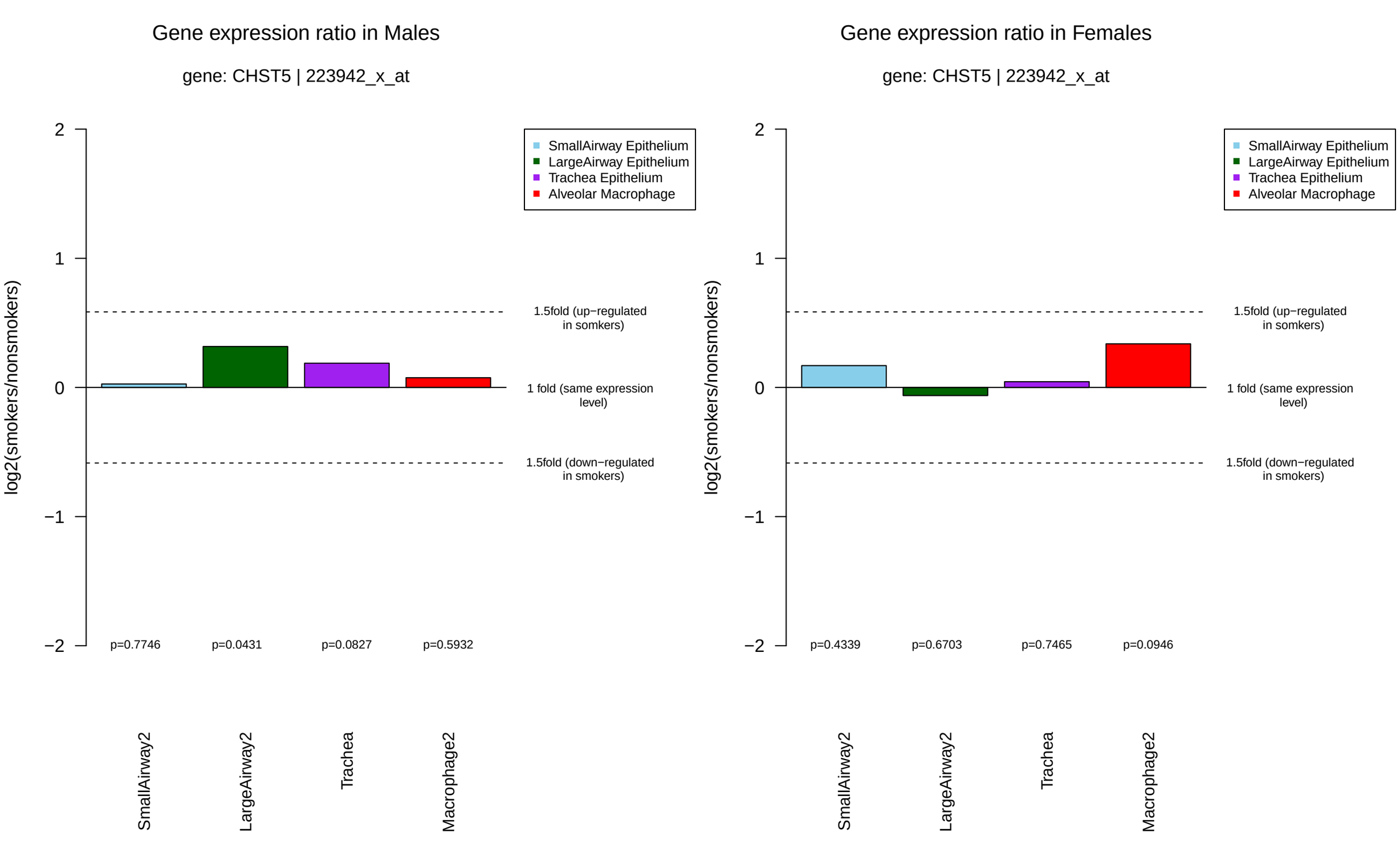
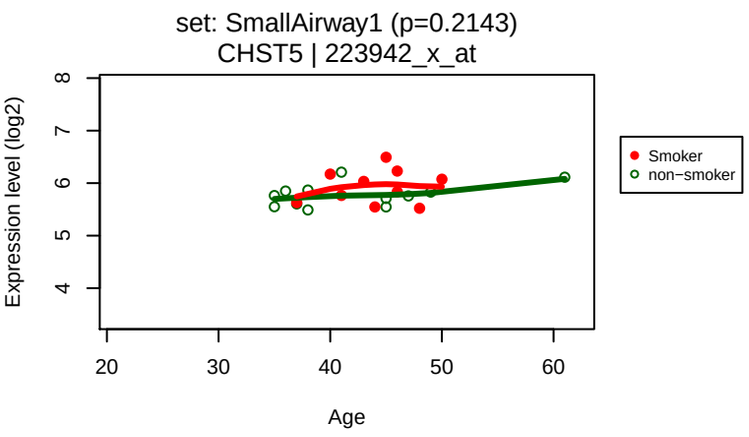
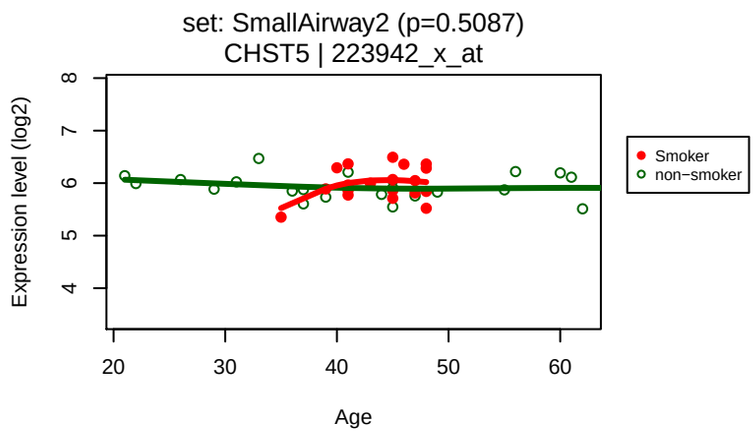
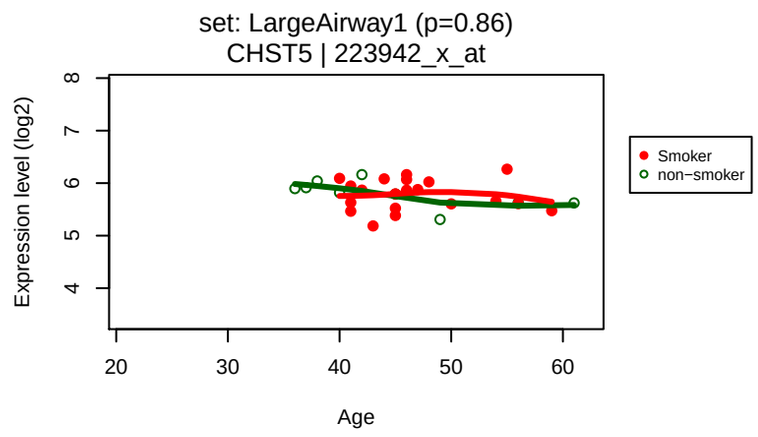

Gene expression of smokers and nonsmokers
gene: CHST5 | 223942_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2143
p=0.5087
p=0.0536
p=0.86
p=0.1157
p=0.1014
p=0.797
p=0.1716
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

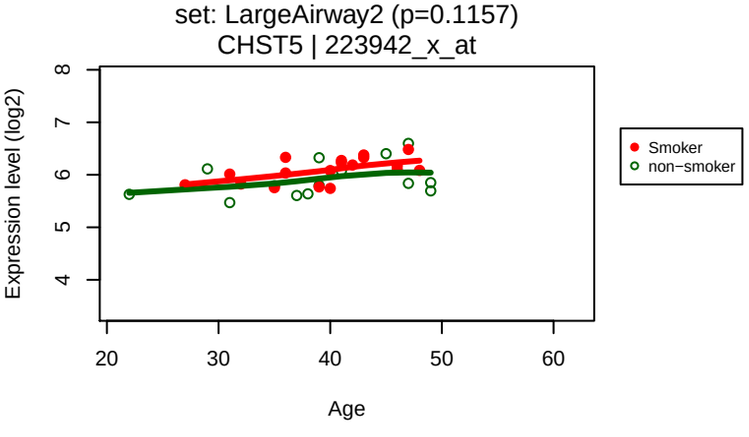

Gene expression ratio (smokers/nonsmokers)
gene: CHST5 | 223942_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2143
p=0.5087
p=0.0536
p=0.86
p=0.1157
p=0.1014
p=0.797
p=0.1716
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

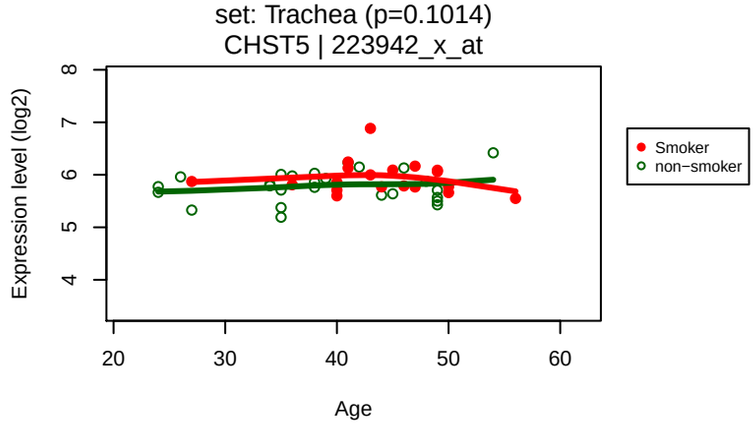

Gene expression ratio in Males
gene: CHST5 | 223942_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7746
p=0.0431
p=0.0827
p=0.5932
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHST5 | 223942_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4339
p=0.6703
p=0.7465
p=0.0946
SmallAirway2
LargeAirway2
Trachea
Macrophage2

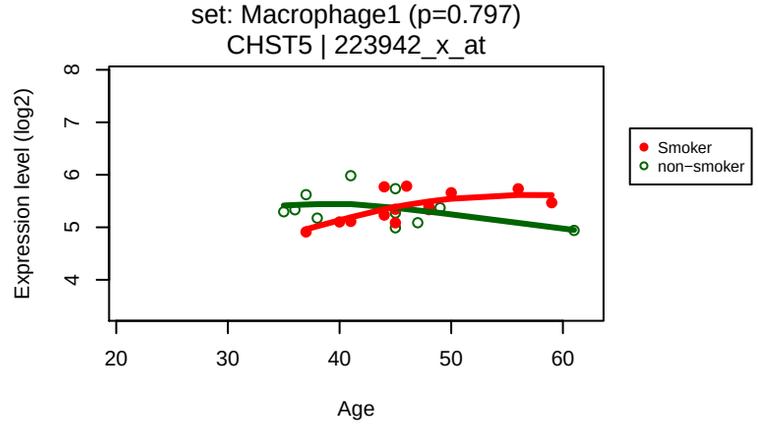

set: SmallAirway1 (p=0.2143)
CHST5 | 223942_x_at
Expression level (log2)
Age
Smoker
non−smoker

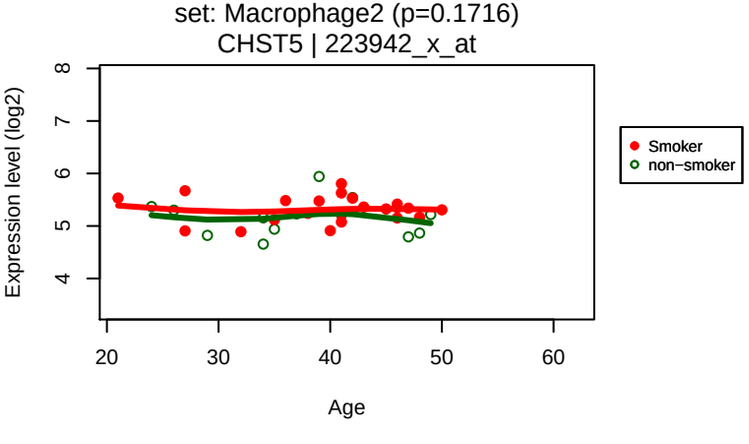

set: SmallAirway2 (p=0.5087)
CHST5 | 223942_x_at
Expression level (log2)
Age
Smoker
non−smoker

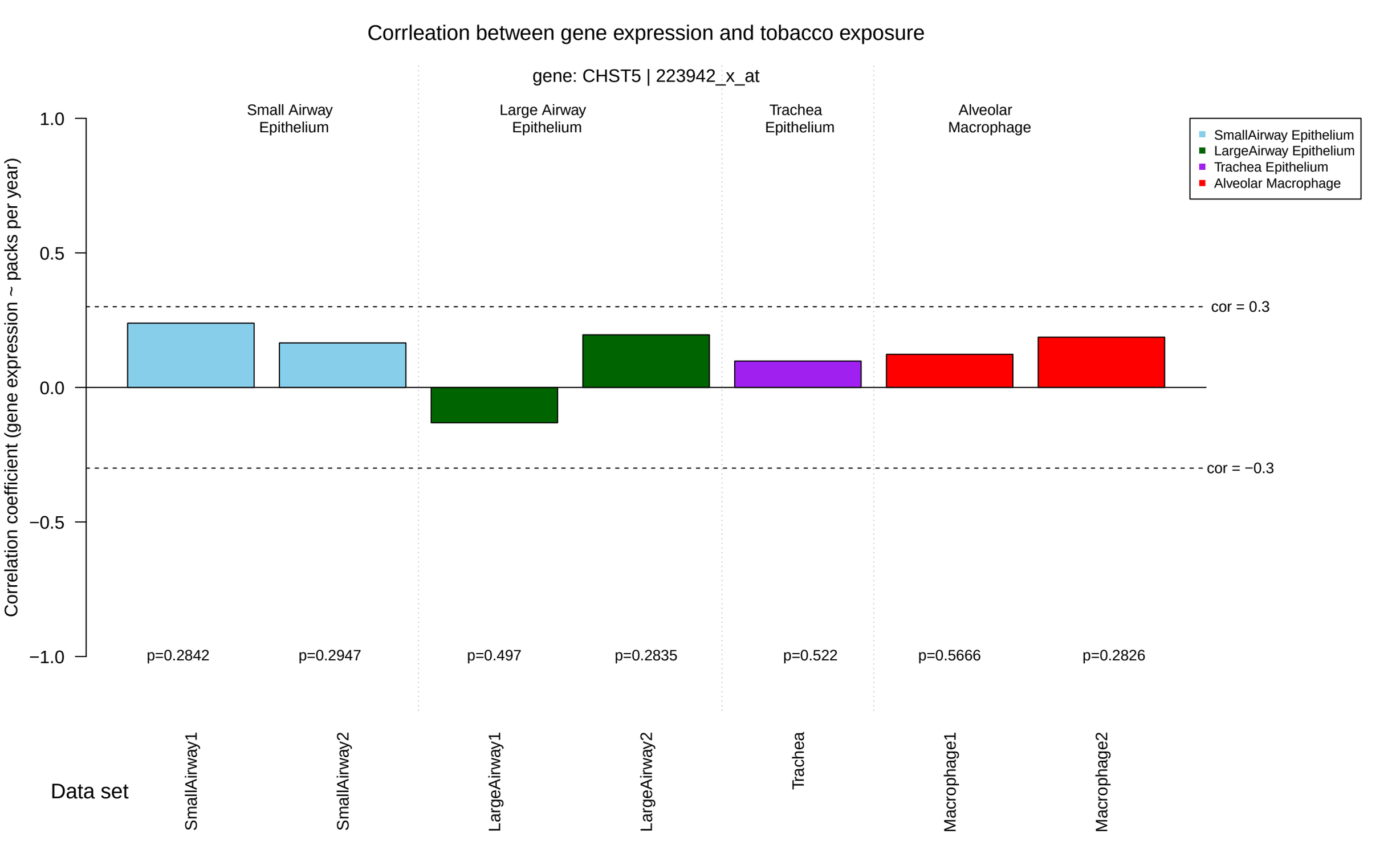

set: LargeAirway1 (p=0.86)
CHST5 | 223942_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1157)
CHST5 | 223942_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1014)
CHST5 | 223942_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.797)
CHST5 | 223942_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1716)
CHST5 | 223942_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHST5 | 223942_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2842
p=0.2947
p=0.497
p=0.2835
p=0.522
p=0.5666
p=0.2826
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2